

UDK Phase 0: Baseline Specification

Sep 29, 2026 · @Sohan

1. Executive Summary

The baseline is GitHub main at commit 96ea038: an English-only speech pipeline that catches 78/84 prepped danger runs, alerts on 26/162 prepped controls (UDK path; the scream trigger was not in that run), and gives 2.2 speech false alarms per hour (4.6 including scream) on 12 h of sealed real audio. Each number is traced to code, recounted from saved run logs, or attributed to the status doc where only it holds the figure; anything that could not be checked is marked NOT VERIFIED.

What it does. Audio arrives over a WebSocket and is cut into utterances by webrtcvad. Each utterance is transcribed by Whisper small.en and Parakeet TDT 0.6B v2, both English-only. Each transcript is matched against 20 fixed phrases plus one personal phrase (exact, fuzzy, semantic, whole segment and per sentence), and the stronger decision wins. A fine-tuned mDeBERTa can silence a standard-phrase alert only when both engines heard the same words and it scores them below 0.02. Separately, BEATs raises "Are you safe?" on a sustained scream or cry.

Key findings.

- **Deployment:** 96ea038 is the tip of main and its config matches the status doc. That AWS runs this exact image is NOT VERIFIED from here (deploy-on-push by design; no AWS access in this audit).
- **Language:** language detection and Indian-language STT exist in code but are off. All speech is treated as English; on real Indic speech (FLEURS, 7 languages) Whisper outputs English text for 96–99% of clips.
- **Metrics reproduce:** 78/84 and 26/162 recount exactly from the 96ea038 run logs. The older 82/84 and 35/96 belong to earlier configurations and a different control count.
- **Main recall gap:** overlapping speech. One two-speaker clip is missed in every audio condition, and synthetic overlapping mixes are caught 1/5.
- **Main false-alarm source on unseen audio:** the scream trigger (2.4/h sealed, 18.7/h in gameplay audio).
- **Operational risks found in code:** a silent placeholder personal phrase is active when none is set; neither the retention sweep nor in-memory journey eviction is ever called; the VAD re-scans the whole journey buffer on every frame; Python packages and most

model revisions are unpinned.

- **Evaluation vs live:** the evaluation harness differs from `api.py` in four documented ways (section 14), so offline numbers approximate, not replicate, live behaviour.

2. Current Production Version

The deployment commit is 96ea038 (2026-09-26 13:29 +0530), the tip of `origin/main`; the local checkout is a different, undeployed branch.

Item	State (checked 2026-09-29)
Repository	<code>github.com/sohans1092004/model</code>
Deployment branch / commit	<code>main @ 96ea038</code> "Deploy: smoke_test.py..." (also branch <code>vad-fragment-fix</code>)
Local branch	<code>intent-context @ a530fa8</code> : 7 commits ahead of <code>main</code> (<code>intent v2</code> , context switches, sealed-test mode). Touches <code>intent_gate.py</code> , evaluation scripts and data; <code>api.py</code> and <code>udk_engine.py</code> unchanged
Uncommitted tracked changes	<code>.gitignore</code> , <code>evaluate_real_world.py</code> ("Files alerting" column), <code>sealed_runs.md</code> (run 1 results)
Other remote branch	<code>decision-logging @ 436246f</code> : alert-reason logging, not merged
Deploy mechanism	Push to <code>main</code> → <code>sakhi-udk-ci</code> builds the image → <code>remote_deploy.sh</code> restarts the container on EC2 <code>c7i.2xlarge</code> (CPU)
Config source	<code>deploy/.env.example</code> overrides SSM for detection flags; <code>UDK_DEVICE=cpu</code> and secrets come from SSM <code>/sakhi/udk/config</code>
Docker	<code>pytorch/pytorch:2.5.1-cuda12.4-cudnn9-runtime</code> , <code>uvicorn api:app</code> , 1 worker, port 8000, volume <code>/data</code>
API entry	<code>api.py</code> : POST <code>/v1/journeys/start</code> , <code>/stop</code> , GET <code>/events</code> , <code>/audio-url</code> , <code>/audio</code> , <code>/metrics</code> , WS <code>/stream</code>

Item	State (checked 2026-09-29)
Offline evaluation entries	evaluate_prepped_data.py, evaluate_full_system.py (real_recordings), evaluate_real_world.py (long harmless audio)
Unit tests	test_*.py (stubbed models); a Colab live-server rehearsal (colab_api_check.py) was run before the merge

Documentation vs repository

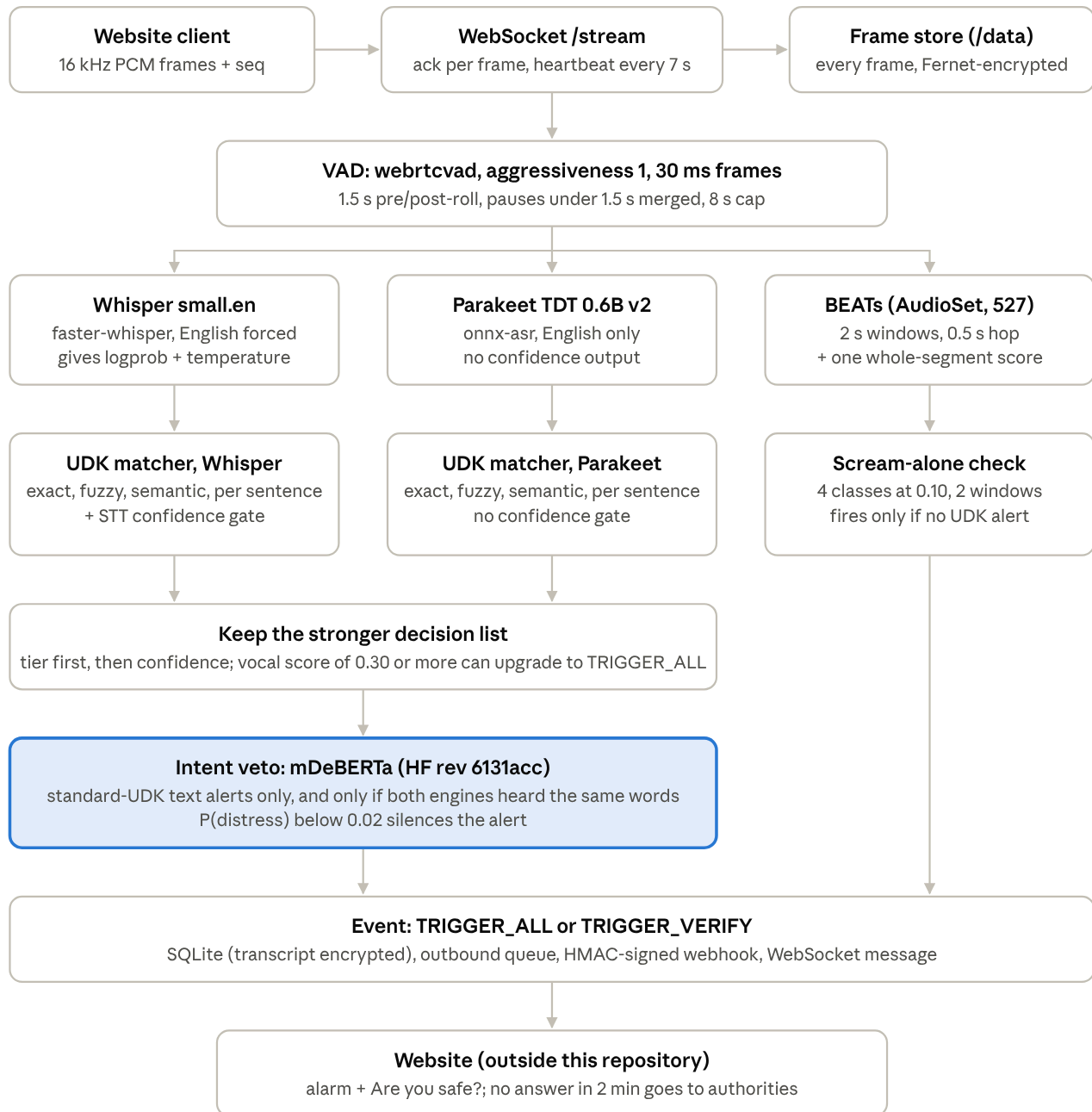
Component	Expected from the status doc	Actual repository state (main 96ea038)	Verified?	Notes
Production commit	96ea038, closed 2026-09-26	origin/main = 96ea038	Yes (repo)	Live AWS image not checked: NOT VERIFIED
STT	Whisper small.en + Parakeet, either alerts	UDK_ENABLE_STT=1, UDK_STT_MODEL=small.en, UDK_ENABLE_PARAKEET=1	Yes	Doc's section 6 table still says Parakeet is "opt-in"; the code default is off, the deploy file turns it on
Keyword spotting	Off	UDK_ENABLE_KWS=0	Yes	The wav2vec2 model is still downloaded into the image
Semantic layer	MiniLM, threshold 0.65	paraphrase-multilingual-MiniLM-L12-v2, 0.65	Yes	api.py comment still says 0.55 (stale comment)
Fuzzy rule	Long sentences need 75% of the UDK's words	FUZZY_MIN_WORD_COVERAGE = 0.75	Yes	
Fragment rule	Short transcripts need an anchor word	_has_anchor(); UDK_FRAGMENT_FUZZY default on	Yes	
VAD	Pauses under 1.5 s stay together	post_roll_ms = 1500	Yes	Also an 8 s maximum segment, not in the doc
Intent veto	mDeBERTa, both STTs agree, $P < 0.02$	intent_gate.py THRESHOLD 0.02, heard_by_both ≥ 80	Yes	Weights pinned to HF revision 6131acc0 in remote_deploy.sh
Scream trigger	On, 4 classes	UDK_ENABLE_SCREAM_TRIGGER=1; Screaming, Crying/sobbing, Whimper, Children shouting	Yes	
Scream corroboration	Diagram: "confirms a weak match"	UDK_ENABLE_SCREAM_DETECTION=1: upgrades an existing alert to TRIGGER_ALL when the 9-class vocal score ≥ 0.30	Partly	Uses the broad 9-class set, not the 4 live trigger classes
STT confidence gate	Not described	UDK_ENABLE_STT_CONFIDENCE_GATE=1: Whisper avg_logprob < -1.0 or temperature > 0 caps a text match below the alert floor	Yes	Missing from the doc

Component	Expected from the status doc	Actual repository state (main 96ea038)	Verified?	Notes
Language ID / Indic STT	English only (doc)	UDK_ENABLE_LANGUAGE_ID and UDK_ENABLE_INDIC_STT unset	Yes	
LaBSE multilingual matcher	Not mentioned	Loaded whenever UDK_ENABLE_SEMANTIC=1, used only after a language switch (never, with LID off); not prefetched in the image	NOT VERIFIED	Whether it downloads and loads in production is unknown; if it does, it costs memory for nothing
Delivery	WebSocket + signed webhook	webhook_delivery.py HMAC-SHA256 when UDK_WEBHOOK_URL is set	Yes (code)	Webhook URL set in SSM: NOT VERIFIED
Rate limit	Not in doc	UDK_RATE_LIMIT_PER_MIN=600 per service token	Yes	

3. Exact End-to-End Architecture

Traced from api.py `_run_incremental_detection` at 96ea038. Everything runs per VAD segment; the only cross-segment state is the matcher's 15 s repetition window and the journey's heartbeat timer.

Two English speech engines vote; BEATs adds scream-only alerts



live inference path in api.py at 96ea038

The highlighted veto is the only component that can remove an alert; the scream path can only add one.

Stage	Input → output	Model / version	Threshold / setting	Enabled in prod	Failure behaviour
Transport	Binary frames: 8-byte seq + 8-byte client time + PCM → ordered buffer	FastAPI WebSocket	Out-of-order frames held until the gap fills; duplicates ignored	Yes	A permanent gap stalls processing after it (no timeout on a missing seq)
Durability	Each frame → encrypted file before processing	Fernet (MultiFernet)	–	Yes	–
Buffer	Whole journey kept in RAM (audio_buffer)	–	No cap	Yes	Grows with journey length; never evicted (section 18)
VAD	Buffer → speech segments	webrtcvad	Aggressiveness 1, 30 ms frames, 1.5 s pre/post-roll, 8 s cap	Yes	Re-run over the whole buffer on every frame; a segment is processed once audio exists after its end
STT 1	Segment PCM → text + avg_logprob, temperature	faster-whisper small.en (revision unpinned)	language="en", library defaults	Yes	CUDA failure falls back to CPU; load failure falls back to MockSTT (empty text), silently
STT 2	Segment PCM → text	Parakeet TDT 0.6B v2 via onnx-asr 0.12.0	English	Yes (English journeys only)	Load failure = None, silently
Scream trigger	Segment → fired/peak/class	BEATs iter3 AS2M cpt1 (lpepino/beats_ckpts)	max of 4 classes ≥ 0.10 in 2 consecutive 2 s windows	Yes	–
Scream score	Segment → max of 9 vocal classes	same BEATs	≥ 0.30 corroborates	Yes	Runs only if some text or scream exists
Confidence gate	Whisper metadata → suspect flag	rule	avg_logprob < -1.0 or temperature > 0	Yes (Whisper text only)	Caps confidence at 0.59
Matcher	Text → decision list (whole segment + per-sentence extras)	rapidfuzz + MiniLM-L12 (revision unpinned)	see section 7	Yes	–

Stage	Input → output	Model / version	Threshold / setting	Enabled in prod	Failure behaviour
Combine	Two lists → the stronger	rule	tier, then confidence	Yes	Engine state kept only for the winning list
Intent veto	Winning list + other engine's text → list	mDeBERTa-v3-base fine-tuned, HF rev 6131acc0	heard_by_both ≥ 80; P < 0.02	Yes, if the model loaded	Missing model: warning, gate off
Scream-alone	No UDK alert + scream fired → TRIGGER_VERIFY	–	–	Yes	–
Event	Decision → event dict	–	event_id = sha256(journey, segment start, UDK)	Yes	Delivery failure stays queued; retried on each heartbeat
Delivery	Event → webhook POST + WebSocket JSON	urllib, HMAC-SHA256	5 s timeout	Yes, if UDK_WEBHOOK_URL set	–
Timeout	No heartbeat or frame	–	20 s grace, 10 s after any alert	Yes	Checked lazily when the journey is next accessed

Latency. Measured on a T4 GPU in the evaluation harness: 0.57 s per prepped clip (status doc; log timings 0.3–1.1 s per clip). A segment can only be processed once 1.5 s of post-roll has arrived after the last word. Production CPU latency (c7i.2xlarge): NOT VERIFIED; a laptop-CPU Docker run needed 1.3–1.7× the audio's length.

4. VAD / Segmentation

Segmentation is webrtcvad at aggressiveness 1 with 1.5 s padding on both sides, pauses under 1.5 s kept inside one segment, and a hard 8 s cap (vad.py at 96ea038). The doc's "pauses under 1.5 s stay together" is verified; the 8 s cap is not in the doc.

Setting	Value (code)	Notes
Library	webrtcvad (webrtcvad-wheels, unpinned)	Frame-level speech/non-speech, no speaker information

Setting	Value (code)	Notes
Speech threshold	Aggressiveness 1 of 0–3	Deliberately permissive: ambiguous audio counts as speech
Frame	30 ms, 16 kHz, 16-bit mono	
Minimum speech duration	None: a single speech frame opens a segment	
Pre-roll / post-roll	1,500 ms / 1,500 ms	
Pause merging	A gap shorter than the post-roll does not end the segment (2026-09-26)	
Maximum segment	8,000 ms of frames from the first speech frame, then a forced cut	After a forced cut the next segment's pre-roll may overlap back; otherwise segments never overlap (2026-09-25 fix)
Multiple speakers	Allowed: one segment can hold several speakers and sentences	Per-sentence matching and both STTs see the mixed segment
Discarded audio	Frames outside segments (beyond the 1.5 s padding) never reach STT or BEATs	Stored encrypted, but never analysed
Timestamps	start_ms/end_ms per segment; event time = journey start + segment start	Word-level timestamps are not used
Live vs evaluation	Same VAD class. Live re-runs it on the whole growing buffer at every frame and processes a segment once audio exists past its end. Evaluation runs it once per file; evaluate_real_world.py per 5-minute chunk	Equivalence of the two NOT VERIFIED; chunking can cut an utterance at a chunk boundary

Measured or observed consequences

Risk	Behaviour at 96ea038	Evidence
Cutting a sentence in half	Only at the 8 s cap, or at a pause of 1.5 s or more	Before the 2026-09-26 fix every negation clip split into "I don't" and "want to go home now"; negation clips alerting went 18/36 → 1/36 after it
Combining unrelated sentences	Yes: speech with pauses under 1.5 s runs together for up to 8 s	"Let go! ... You're hurting me!" is one segment, which is why per-sentence matching exists; the intent veto judges whole-segment text
Losing short cries	Not observed on prepped; a cry web rtcvad does not flag as speech is never analysed, by STT or by BEATs	Not measured
Losing quiet second-speaker speech	The VAD keeps it; the loss happens in STT (section 5)	"Exact UDK_03 + UDK_09" missed in all 6 conditions
Creating fragments	Short segments still arise from isolated words	Fragment and filler rules exist because of them (section 7)
Duplicate transcription	Fixed 2026-09-25: one "Please" was transcribed twice and escalated as a repeat	Status doc section 16 / vad.py comment
Changing context	Context available to any model = one segment of 8 s or less	Relevant to section 11

5. ASR Pipeline

Both deployed speech models are English-only, and nothing in production detects or switches language. Their outputs are combined by keeping whichever transcript's decision is stronger (an OR), not by voting.

Property	Whisper (primary)	Parakeet (second)
Model	faster-whisper <code>small.en</code> (an English-only checkpoint)	NVIDIA Parakeet TDT 0.6B v2, ONNX via onnx-asr 0.12.0
Checkpoint pin	None: downloaded by name at image build	None: <code>nemo-parakeet-tdt-0.6b-v2</code> fetched at build
Language	Forced: <code>transcribe(audio, language="en")</code>	English only (model)
Decoding	Library defaults (beam 5, temperature fallback 0.0→1.0; unpinned library version, so defaults may drift)	Model default (TDT greedy)
Built-in VAD filter	Not used (<code>vad_filter</code> default off)	–
Timestamps	Segment-level from the library; not used downstream	Not used
Confidence	<code>avg_logprob</code> , temperature, compression ratio, no-speech probability	None
Hallucination guard	STT confidence gate: <code>avg_logprob < -1.0</code> or temperature <code>> 0</code> caps the text match at 0.59 (below the alert floor). Calibrated on <code>n = 8</code> (<code>stt_confidence_gate.py</code> says "not final")	None
Matching rules applied	Full matcher, per sentence, plus the confidence gate	Same matcher and per-sentence pass, no confidence gate
Device	CUDA if configured, else CPU int8; production CPU	CPU in production
Latency	About 4.2 s per 5 s clip on CPU (status doc, older measurement); part of 0.57 s per clip on T4	About 1.1 s per clip on CPU (status doc)
Memory	NOT MEASURED	Model about 2.4 GB on disk; RAM NOT MEASURED

Property	Whisper (primary)	Parakeet (second)
Code-switching	Not supported: Indic speech comes out as English words (section 6)	Same
Output normalization	Lower-case, punctuation and symbols stripped before matching; punctuation first used to split sentences	Same

How the two are combined (api.py `_either_stt_decisions`): the matcher decides on Whisper's text, then re-decides the same segment from the same engine state on Parakeet's text, and keeps the list whose best decision is stronger (tier, then confidence). The engine's repetition memory keeps only the winning list, so two transcripts of one utterance never count as a repeat. Parakeet's text is also the "other engine" text the intent veto compares against (or Whisper's, when Parakeet's list won).

Measured contribution (prepped, earlier configurations): Whisper alone 78/84 vs Whisper + Parakeet 82/84 danger runs; Parakeet caught loud, crowd and pocket runs Whisper broke on, while Whisper caught a short quiet second-speaker phrase Parakeet dropped. On real speech the OR added false alarms: 42 vs 33 per hour before the fuzzy fix. At 96ea038 on dev real-world audio: Whisper-only 3.1, Parakeet-only 1.3, either 3.6 alerts per hour (conversation).

Observed hallucinations: "Thanks for watching!" on traffic noise; on non-English FLEURS speech, YouTube-outro text such as "Please remember to subscribe to our channel" (section 6).

6. Multilingual / Code-Switching Status

The deployed system has no multilingual path: Indian-language speech is transcribed as invented English, and danger recall in any Indian language has never been measured. What has been measured is how often that invented English triggers alerts.

Language / mix	Danger recall	False alarms	Source
English	78/84 prepped runs (scripted, 6 conditions); 3/7 real media clips	26/162 prepped controls; 2.2/h sealed real audio	Section 14
Hindi	Not measured	Not measured on real Hindi conversation	–
Telugu	Not measured	4.2/h on own recordings (1.2 h, 4 files) at 96ea038; a 2026-09-27 run of the same files reported 5.1/h (unexplained difference, both recorded)	vad_fix_results.zip; memory of 2026-09-27 run
Hinglish	Not measured	Not measured	–
Telugu-English	Not measured	Included in the 4.2/h above (English words inside Telugu speech)	–
Other mixed	Not measured; 3/7 real distress media clips (Hindi, Tamil, Malayalam-dubbed films) were caught, each via English words	1/23 real media negatives alerted (an English podcast)	real_recordings

Code exists but is disabled: Whisper/VoxLingua language ID, fine-tuned Indic Whisper models (te, kn, hi, ta), Indic keyword spotting and LaBSE matching. The code comments quote 95–100% accuracy for those Indic models; the evaluation data behind those figures was not found in this audit (NOT VERIFIED), and none of it is live.

A. ASR failure: Indian speech becomes English text

Both live STTs were run on FLEURS dev speech (real read speech, 7 Indian languages, about 300 clips each) during intent v2 work. Whisper returned English text for nearly every clip:

Language	Clips	Whisper non-empty	Parakeet non-empty	Transcripts that would raise a UDK (exact/fuzzy only)
Telugu	300	298	293	7 of 591
Hindi	239	227	230	7 of 457
Tamil	300	287	286	8 of 573
Bengali	300	289	296	6 of 585
Kannada	299	289	288	5 of 577
Marathi	299	288	289	7 of 577
Malayalam	300	282	298	1 of 580

Examples: Telugu → "Not only the platelet chappan walk Hindu matam laya..."; Bengali → "Please remember to subscribe to our channel...". The UDK column undercounts: the harvest had no semantic layer loaded.

B. Matcher failure caused by ASR

Real Telugu recordings, dev alerts: "What's the name of the police?" → UDK_11 (semantic); "I need that to rewrite it." → UDK_20 (fuzzy); "Ah yeah, uppartana... let go of me" → UDK_10; "Free trades are there.". On the 2026-09-27 dev review, 20 of 26 unique speech false alarms (YouTube and Telugu) were heard by only one engine, the signature of invented text.

C. Context-classifier failure caused by ASR

The live veto judges only words both engines heard, so it never sees most invented-English alerts: it neither silences nor worsens them. Letting it judge single-engine text was tested and rejected (it vetoed a real cry). Beyond that: not measured.

7. UDK Matching System

Every transcript is scored against all 21 phrases (20 standard + 1 personal) by three layers, and the single highest-confidence match wins (`udk_engine.py match_transcript`). All four doc claims check out in code: the 75% word rule, the fragment rule, keyword spotting off, and different treatment of confident vs weak matches. One behaviour is not in the doc: the semantic layer ignores the fragment rule.

Before any layer: text is lower-cased and punctuation/symbols are stripped. A transcript made only of filler words (please, now, right, okay, I, me, you, the...) never matches.

Layer	Algorithm	Fires when	Confidence given
Exact, whole phrase	Substring	The UDK phrase appears inside the transcript	1.0
Exact, fragment	Substring the other way	The transcript appears inside a UDK phrase AND contains one of its anchor words (non-function words; "don't" counts only as a command)	1.0, or 0.65 if the fragment fits inside more than one UDK
Fuzzy	rapidfuzz partial_ratio and token_set_ratio	Both ≥ 0.55 , AND a long transcript contains $\geq 75\%$ of the UDK's words (each word $\geq 75\%$ similar) within a phrase-length window, AND a short transcript has an anchor word	$\text{partial_ratio} \times 0.9$ (capped at 0.65 for ambiguous fragments)
Semantic	Cosine similarity, paraphrase-multilingual-MiniLM-L12-v2	≥ 0.65	The cosine itself, not discounted
Keyword spotting	wav2vec2 + DTW against TTS references	Disabled (UDK_ENABLE_KWS=0)	0.65

Layer	Algorithm	Fires when	Confidence given
Per sentence	Whisper/Parakeet punctuation splits sentences	A sentence matches a different UDK at ≥ 0.60 ; sentences under 3 words count only if they appear word for word in a UDK	Its own decision, reported as an extra event
Custom phrase	Same three layers as the standard phrases	–	–

The semantic layer applies no anchor or filler check beyond the all-filler rule, so short fragments can still match by meaning: "Goodbye. Goodbye." scored as UDK_10 ("Let go of me") on a real media clip (the veto then silenced it).

When a match becomes an alert

Condition (after the best match is chosen)	Result
No match, or confidence < 0.60	NO_ACTION
Whisper transcript flagged suspect (avg_logprob < -1.0 or temperature > 0) and no keyword-spotting agreement	Confidence capped at 0.59, so NO_ACTION
UDK_14, UDK_18 or UDK_20 ("verify by default"), not repeated and not corroborated	TRIGGER_VERIFY
Confidence ≥ 0.85	TRIGGER_ALL
Same UDK again within 15 s, or 2 different UDKs within 15 s	TRIGGER_ALL
BEATs 9-class vocal score ≥ 0.30 on the segment	TRIGGER_ALL
Keyword spotting agrees (distance ≤ 0.15)	TRIGGER_ALL (inactive: KWS off)
Otherwise, $0.60 \leq \text{confidence} < 0.85$	TRIGGER_VERIFY

Then, across the two engines, the stronger decision list is kept, and the intent veto may silence it (section 12). Every alert, of either tier, rings the alarm and asks "Are you safe?" on the website, so the tier affects wording, not whether the user is alerted.

8. Custom UDK System

A journey can carry one personal phrase, sent as text at journey start; when none is sent, a fixed placeholder phrase is silently active for that journey. Both confirmed in code at 96ea038 (api.py `start_journey`, udk.py line 239, enrollment.py).

Question	Current behaviour
How provided	<code>personal_udk_phrase</code> in POST <code>/v1/journeys/start</code> (text only)
Validation	Not empty; at least 3 words; <code>token_set_ratio</code> < 0.75 against 9 hard-coded ordinary sentences ("call me later", "let's get out of here"...)
Maximum length	None
Number of phrases	One per journey
Where stored	Journey row in SQLite at <code>/data</code> (or Postgres if <code>UDK_DATABASE_URL</code> is set; not set in the deploy file)
Encryption	Field-level Fernet on <code>personal_udk_phrase</code>
ID	<code>PUDK_<user_id></code>
Matching	Same exact, fuzzy, semantic and per-sentence layers as the standard phrases, on both engines
Special treatment	Type PERSONAL: never judged by the intent veto; <code>verify_by_default</code> off, so a confident match is a full TRIGGER_ALL
Voice enrollment	<code>enroll()</code> can take a sample and check intelligibility, but the API never passes audio: not available
Keyword spotting	Not available (needs a voice sample, and KWS is off)
Placeholder	"the weather in Denver is lovely" (ID PUDK_U1001) is active whenever no phrase is given , for every such user, with semantic matching. Status doc section 1 lists it; it still exists

Question	Current behaviour
Evaluated?	No: every evaluation script builds its engine from the 20 standard phrases only, so neither custom phrases nor the placeholder have measured false-alarm or recall numbers

9. Acoustic Distress System

One BEATs model does two different jobs with two different class sets: a 4-class scream trigger that can raise "Are you safe?" on its own, and a 9-class score that upgrades an existing UDK alert to TRIGGER_ALL. The live trigger classes are confirmed as Screaming, Crying/sobbing, Whimper and Children shouting.

Property	Scream-alone trigger (scream_trigger.py)	Corroboration score (udk_engine.py)
Model	BEATs_iter3_finetuned_on_AS2M_cpt1.pt from HF lpepino/beats_ckpts (revision unpinned)	same instance
Input	VAD segment, 16 kHz; segments under 200 ms score zero	whole VAD segment in one pass
Classes	Screaming, Crying/sobbing, Whimper, Children shouting	Screaming, Yell, Shout, Children shouting, Battle cry, Crying/sobbing, Baby cry, Whimper, Groan
Window / aggregation	2 s windows every 0.5 s; max of the 4 classes	max of 9 classes over the segment
Threshold	≥ 0.10 in 2 consecutive windows (about 2.5 s of audio)	≥ 0.30
Effect	TRIGGER_VERIFY with layer <code>acoustic_scream</code> , only when the segment has no UDK alert	Lifts an existing alert, including verify-by-default phrases, to TRIGGER_ALL; never creates one
Production	On (UDK_ENABLE_SCREAM_TRIGGER=1)	On (UDK_ENABLE_SCREAM_DETECTION=1)
Limit	Audio outside VAD segments is never scored	same

Measured results

Data (type)	Result	Configuration
FSD50K real screams (real, calibration set)	11/25 caught	9-class rule, 2026-09-24
Real crying (calibration)	16/25	9-class rule
Real screams + cries + distress media, 57 clips	29/57 caught (9 classes: 31/57)	4-class rule
Real distress media, 7 clips	Scream alone 4/7; with UDKs 6/7	9-class rule
Real Indian YouTube, 4.6 h, harmless (dev)	182 alerts, 20–77/h	9 classes; causes Battle cry 65, Wail/moan 69, Groan 34, Screaming 10
Same audio (dev)	Hard audio 7.3/h (gameplay 18.7, festival 1.8), conversation 1.8/h, ambient 0	4-class rule, 96ea038
Telugu own recordings, 1.2 h (dev)	0/h	4-class rule
Sealed real YouTube, 12 h	2.4/h	4-class rule, 96ea038
Real 911 calls, 220 dev calls (partial run)	11.9 alert episodes/h; not scored per scream, so true vs false unknown	4-class rule

Environmental false-alarm sources: recorded screams in video-game audio (all 10 live-rule dev alerts came from one gameplay video); cheering, crowd and laughter added 1–3 of about 25 clips each in calibration. Sirens: 0 with BEATs (emotion2vec fear, rejected, fired on 12–14 of 25).

Missed distress: frightened speech barely registers (BEATs 0.013–0.022 on the "Real fear, in pocket" clip); 28 of 57 real distress sounds are missed by the live rule.

Rejected changes: a music/game/TV/radio veto (dropped real distress media 3/7 → 1/7); 3–4 consecutive windows (lost screams 11 → 8 and cries 15 → 13); emotion2vec fear as a trigger; weak-UDK suppression (lost 2 real detections). Untested: a veto on "Video game music" alone.

10. Intent / Context Models

Only one intent model is live: a fine-tuned mDeBERTa used as a veto on standard-UDK text alerts. Every other model was an experiment and was rejected, most of them for losing real danger detections.

Shared across all of them: a 2-way NLI setup with the hypothesis "The speaker is in immediate danger or is asking for help." (positive = distress, negative = not distress). None uses speaker identity, timestamps or language information. Training data is generated text (intent_data/generate.py templates, later LLM prompt outputs and FLEURS STT mistakes), split by template.

Model	Status	Training data	Context	Threshold	Test result (generated set)	On real prepped transcripts	Why rejected
nli-MiniLM2-L6 zero-shot	Rejected	none	sentence	0.04	recall 99.5%, FP 91%	gate: lost 0, controls 46/96	Flags everything
nli-deberta-v3-small zero-shot	Rejected	none	sentence	0.01	81% / 73%	lost 2, controls 38/96	Poor separation
deberta-v3-base zero-shot	Rejected	none	sentence	0.01	61% / 14%; 1,013 ms per window on CPU	lost 9	Misses danger, slow
Qwen2.5-1.5B / Phi-3.5-mini zero-shot	Rejected	none	sentence	-	Qwen flagged 38% of negations; Phi missed 22% of danger	-	Both failure types
MiniLM + logistic regression	Rejected	5,388 generated sentences	sentence	0.66	92.5% / 1.9%	alone 71/84 danger; gate lost 8	All 8 losses were garbled transcripts
deberta-v3-small fine-tuned	Rejected 2026-09-25	4,504 train / 1,126 dev / 957 test	sentence	0.99	98.0% / 4.2%; 19 ms GPU	alone 75/84, controls 2/96; gate lost 1	Silenced a garbled danger clip at every threshold
mDeBERTa-v3-base fine-tuned, round 1	Superseded	same	sentence	0.99	97.5% / 3.6%; 27 ms GPU	alone 80/84; gate lost 3	-

Model	Status	Training data	Context	Threshold	Test result (generated set)	On real prepped transcripts	Why rejected
mDeBERTa v1 (negation retrain)	Production since 2026-09-26	round 1 + about 2,700 negation minimal pairs + 226 negation-but-danger sentences	segment transcript (the whole VAD segment or one sentence)	0.02 veto, only when both STTs agree	0/220 negations flagged, 12/12 negation-but-danger kept; threshold keeps 99.7% of test distress	Prepped 78/84 kept, controls 26/162 at 96ea038	-
mDeBERTa v2 (context)	Rejected 2026-09-28	intent_dataset_v2.csv: 18.5k rows incl. 2,000 LLM lines and FLEURS mistakes	previous 20 s of transcript prepended	0.05	dev loss 0.0285	78/84 (tie)	Tied on sealed audio; silences cries after an attacker's words (section 11)

Production CPU inference time for the live veto: NOT MEASURED. Known failure of the live model: it is badly calibrated on short everyday fragments ("Come on now" and "Tatiana, you want to come in?" score $P(\text{distress})$ 0.999), so it cannot silence those, and its veto is designed never to act on them unless both engines agree.

11. Previous Context Experiment

The v2 context model learned that a context addressing "you" means harmless, because its training data almost never showed an attacker speaking before a cry. That is why it silences real cries after an attacker's words while clearing everyday false alarms. Code is on the undeployed branch intent-context (intent_gate.py, intent_data/finetune.py, build_v2.py).

Question	Answer from the implementation
What counted as context	Transcripts of earlier VAD segments within <code>UDK_INTENT_CONTEXT_S = 20 s</code> , plus the part of the current segment before the alert's sentence
How much	Last 60 words, cut from the front

Question	Answer from the implementation
Before or after the phrase	Before only
Input format	"Before: <context> Now: <alert text>" vs the NLI hypothesis; the sentence alone when there is no context (as in v1)
Speaker identity	None: all speakers' words are merged into one string
Same speaker for context and target	Unknown to the model; attacker and victim words look identical
Text or audio	Text only (the STT transcripts)
Other switches tested with it	One-word hits alert only if $P \geq 0.9$ with context (user rule); veto threshold 0.05; single-STT judging off
Training	mDeBERTa-v3-base fine-tuned on intent_dataset_v2.csv: 18,503 rows, 2,631 with context (1,174 of them positive). Context rows come from 2,000 LLM-written lines (p1-p4)
Evaluation	Dev loss 0.0285; thresholds picked on dev; prepped danger 78/84; sealed 12 h harmless audio (tie with v1); a hand-written probe of 13 sentences

Why it failed: what the data taught

The LLM-written context rows were classified by a simple wording rule (2nd person or imperative with no "he/she/they" = attacker-like; he/she/they = third-person narration). The rule is a heuristic, but the imbalance is large:

Context style	Positive (distress) rows	Negative rows
Third-person narration ("He locked the flat...")	606	47
Attacker-like address or command ("You're coming with me")	7	121
Other wording	88	687

Context style	Positive (distress) rows	Negative rows
No context	99	345

So "you" and commands in the context were mostly paired with harmless labels ("Reverse slowly, I'm guiding you" → "Back!" = 0). That matches the probe exactly: cries after victim narration stayed around 0.99, and cries after attacker speech fell to 0.000.

Failure cases and their likely technical cause

Case	v1 alone	v2 with context	Likely cause
"Get away!" after "You're not leaving!"	0.998	0.000	Attacker-address context learned as harmless; no speaker marking
"Let go of me!" after "You're coming with me whether you like it or not"	0.999	0.000	Same
"Don't touch me!" after "Come here, nobody can hear you"	0.999	0.059	Same; one point above the 0.05 veto
"Help!" alone	0.996	0.001	Training taught that a bare word needs supporting context (the one-word rule's data)
"Get away!" alone	0.998	0.000	Same

Why earlier tests missed it: the dev set came from the same LLM distribution (a 0.0285 dev loss means in-distribution, not robust); the sealed set was harmless-only, so it could measure false alarms but not lost cries; prepped data has almost no cry right after an attacker's line.

12. Final Alert Decision Logic

Per segment, an alert fires if either engine's text reaches confidence 0.60, unless the veto silences it; a scream fires "Are you safe?" only when no UDK alert is left. Traced from `api.py` `_run_incremental_detection`, `_either_stt_decisions`, `_intent_veto` and

udk_engine.py decide .

UDK type	Best match	Engines	BEATs	Intent P(distress)	Result
-	none on either engine	-	4-class rule fired	-	TRIGGER_VERIFY, layer acoustic_scream
-	none	-	not fired	-	Nothing
Standard	confidence $\geq$ 0.60	both heard the same words (token_set_ratio $\geq$ 80)	any	< 0.02	Silenced (then the scream rule may still fire)
Standard	confidence $\geq$ 0.85	both agree	any	≥ 0.02	TRIGGER_ALL
Standard	confidence $\geq$ 0.60	the other engine heard different words, or nothing	any	not judged	Alert stands (VERIFY or ALL by the rules below)
Standard	0.60–0.85	-	vocal score $\geq$ 0.30	≥ 0.02 or not judged	TRIGGER_ALL
Standard	0.60–0.85	-	below 0.30	≥ 0.02 or not judged	TRIGGER_VERIFY
UDK_14 / 18 / 20	≥ 0.60	-	below 0.30, no repeat	≥ 0.02 or not judged	TRIGGER_VERIFY
Any	≥ 0.60 , same UDK within 15 s, or a 2nd different UDK within 15 s	-	-	≥ 0.02 or not judged	TRIGGER_ALL
Personal	≥ 0.60	-	-	never judged	VERIFY (0.60–0.85) or ALL (≥ 0.85 or escalated)
Standard, Whisper text suspect	any	-	-	-	Whisper's match capped at 0.59; Parakeet's list can still alert (no gate on it)

Specific questions

Question	Answer from code
What causes TRIGGER_ALL	Confidence ≥ 0.85 , a repeat or a 2nd distinct UDK within 15 s, or a 9-class vocal score ≥ 0.30 (KWS agreement too, but KWS is off)
What causes TRIGGER_VERIFY	Confidence 0.60–0.85; verify-by-default phrases; a scream alone
What causes silence	No match ≥ 0.60 ; filler-only or anchorless fragments; the confidence gate; the intent veto
What can veto	Only the intent veto (and the confidence gate, on Whisper text)
What cannot be vetoed	Personal-phrase alerts, scream alerts, keyword-spotting alerts, and any alert where Parakeet returned no text or the engines disagree
Whisper and Parakeet disagree	The stronger decision wins; the veto is not applied because the words differ
Both uncertain	Each needs ≥ 0.60 on its own; there is no joint scoring, so two weak signals never add up
ASR hallucination	Caught only if Whisper's own metadata flags it; a single-engine invented phrase can alert and cannot be vetoed
No speech detected	Audio outside VAD segments is never analysed; a segment where both engines return nothing and no scream fires is skipped
Known side effect	A vetoed alert is still counted in the 15 s repetition window, so the next alert in that window is escalated to TRIGGER_ALL (noted in code as accepted)

13. Evaluation Datasets

Only two sets are close to independent tests: the 12 h sealed YouTube audio and the sealed half of the 911 calls. Every other set has shaped design decisions, so its numbers are regression checks, not evidence of generalisation.

Dataset	Type	Size	Danger / controls	Speakers	Language	Seen during development?	Labels
prepped_data	Scripted recordings, plus 5 synthetic audio conditions	41 labelled files × 6 conditions	14 danger / 27 controls (84 / 162 runs)	A handful (a few female and male voices; exact count NOT VERIFIED)	English	Yes, heavily: rules were added after specific misses (per-sentence matching, filler words, the VAD duplicate fix)	From file names, by the author; not independently checked
real_recordings/mixed	Synthetic: TTS UDK phrases mixed with noise	30	30 danger (clean 9, muffled 9, noisy 7, overlapping 5)	TTS voices	English	Yes (calibration)	By construction
real_recordings/positives	Real film and media clips	7	7 danger	Actors	Hindi, Tamil, Malayalam-dubbed, English	Yes (scream calibration set)	Author judgement
real_recordings/negatives	Real podcasts and media	23	23 controls	Many	English, Tamil, Telugu, Kannada, Hindi	Yes (calibration)	Author judgement
FSD50K subsets	Real sound events	about 25 per class (screams, crying, cheering, crowd, laughter, siren, horn)	screams/cries = positives	Many	none	Yes: scream thresholds were picked on them	FSD50K labels
YouTube real_world (dev)	Real Indian YouTube, harmless	4.6 h, 7 files	0 danger	Many	English-heavy, Indian accents	Yes: every alert was reviewed; the fuzzy fix and scream narrowing were designed on it	Harmless by selection
Telugu own recordings (dev)	Real conversation	1.2 h, 4 files	0 danger	Few (NOT VERIFIED)	Telugu with English words	Yes	Harmless by selection
Chess audio (dev)	Real	1.6 h	0 danger	NOT VERIFIED	English	Yes (listed in dev_false_alarms.txt)	Harmless
Sealed YouTube	Real, harmless	12.0 h (25 clips, first 60 min of each)	0 danger	Many	NOT VERIFIED (not inspected by design)	No: used once, totals only	Harmless by selection
911 Recordings (Kaggle)	Real US emergency calls	707 calls: 355 dev / 352 sealed (file-name hash)	Emergencies, plus a few false-alarm calls per metadata	Callers and dispatchers	English	Dev: partially run; sealed: untouched	Dataset author's metadata

Dataset	Type	Size	Danger / controls	Speakers	Language	Seen during development?	Labels
Intent dataset v1	Generated text	6,587 sentences (4,504 / 1,126 / 957)	both	–	English	Split by template; prepped transcripts kept out	By template
Intent dataset v2	Generated text + LLM lines + FLEURS STT output	18,503 rows	both	–	English + invented English	Dev false alarms filtered out	By generator / LLM

Leakage and bias assessment

Risk	Where	Assessment
Tuning against the test set	prepped_data, YouTube dev, real_recordings	High: thresholds and rules were changed after looking at their failures. The 2026-09-28 sealed protocol exists because of this
Speaker leakage	prepped_data	High: few speakers, the same voices across danger and controls, and the same clips in every audio condition (6 runs of one recording are not independent)
Template leakage	Intent datasets	Moderate: splits are by template, but all templates come from one generator, so the test set is "too easy" (status doc section 10)
Scenario leakage	prepped_data	High: 14 danger scenarios, each named after the UDK it contains
Synthetic-data bias	Audio conditions, TTS mixes, generated text	High: augmentation approximates pocket or phone audio; TTS lacks panic; generated contexts lack attacker speech (section 11)
Class balance	Real-world sets	Harmless-only sets measure false alarms but can never show a lost cry

14. Reproduced Baseline Metrics

The 96ea038 prepped numbers reproduce exactly when recounted from the saved run logs (vad_fix_results.zip, Kaggle T4, 2026-09-26): 78/84 danger runs and 26/162 control runs alerting. No model was re-run for this audit (project rule: real-model runs happen on Kaggle/Colab only, with approval).

These prepped and real_recordings numbers are UDK-only: neither harness runs the scream-alone trigger, so live false alarms on those clips could be higher. NOT MEASURED.

Prepped data by audio condition (UDK path, Whisper + Parakeet, intent veto 0.02)

Condition	Danger caught	as TRIGGER_ALL	as TRIGGER_VERIFY	Controls alerting	Missed clip(s)
Normal	13/14	11	2	5/27	UDK_03 + UDK_09 (pocket recording)
Quiet (-12 dB)	13/14	11	2	4/27	same
Loud (+12 dB, clipped)	14/14	10	4	3/27	-
Crowd (babble, 10 dB SNR)	13/14	12	1	3/27	same
Phone (300-3,400 Hz)	13/14	10	3	5/27	same
Pocket (low-pass, -6 dB)	12/14	10	2	6/27	both UDK_03 + UDK_09 recordings
Total	78/84	64	14	26/162	one two-speaker scenario

Controls that alert repeatedly: "Negative - No UDK", "Third-person or quoting a story", "Trap - no UDK - 1" (5-6 conditions each), "Casual chat, overlapping", "Mock-annoyed, overlapping" (4 conditions), "negation_udk1" (pocket only).

real_recordings (same run)

Corpus (type)	Result	Mean latency (T4)
Mixed TTS + noise (synthetic), 30	23/30 caught: clean 8/9, muffled 7/9, noisy 7/7, overlapping 1/5	0.60 s

Corpus (type)	Result	Mean latency (T4)
Real distress media, 7	3/7 caught	8.0 s per clip (clips up to 11 s)
Real ordinary media, 23	1/23 false alarm (fuzzy "...taken away from me" → UDK_03); 1/115 segments	5.7 s per clip

Per-engine and per-path (dev real-world audio, 96ea038, alerts per hour)

Audio	Whisper only	Parakeet only	Whisper + Parakeet	Scream-alone	Everything live
Conversation (2.25 h)	3.1	1.3	3.6	1.8	5.3
Hard: debate, festival, gameplay (2.06 h)	2.9	2.9	4.9	7.3	12.2
Ambient traffic (0.26 h)	0.0	3.9	3.9	0.0	3.9
Telugu own recordings (1.18 h)	3.4	0.8	4.2	0.0	4.2

Whisper-alone and Parakeet-alone recall on prepped at 96ea038: NOT MEASURED (the logs only record the combined result). Earlier configuration: Whisper alone 78/84, Whisper + Parakeet 82/84.

Intent veto contribution

Measure	Without veto	With veto (0.02)	Source
Prepped danger (pre-VAD-fix code)	76/84	76/84 (1 danger sentence vetoed; clip still alerted)	status doc section 16
Prepped controls (pre-VAD-fix)	59/162	43/162	intent_gate.py docstring
real_recordings media negatives	4/23	1/23	status doc section 16
YouTube conversation, speech alerts/h	16.0	4.0 (measured at the old 0.99 threshold)	status doc section 16

Older figures in the status doc

Figure	Status
82/84 danger runs	Earlier configuration (before the fragment rule, veto and VAD fix); not reproducible at 96ea038, which gives 78/84
35/96 controls after the fuzzy fix	Earlier control set of 16 clips $\times$ 6; the set grew to $27 \times 6 = 162$ when negation clips were added. Not comparable
About 16 false alarms per hour on replayed real speech	A transcript replay (2026-09-25), superseded by audio runs: 16.0/h conversation before the veto, 3.6/h at 96ea038
78/84 Whisper-only vs 82/84 Whisper + Parakeet	Earlier configuration; logs for that run not found in this audit (NOT VERIFIED)

Why offline numbers only approximate live behaviour

1. Each engine gets its own matcher in evaluation; live uses one matcher and keeps the stronger list, so escalation to TRIGGER_ALL can differ (alert vs no alert is the same OR).
2. Evaluation matches only the 20 standard phrases; live also matches the personal phrase or the Denver placeholder.
3. Prepped and real_recordings evaluations skip the scream-alone trigger; evaluate_real_world.py adds it.
4. Evaluation segments each file at once (real-world: per 5-minute chunk); live re-segments a growing buffer incrementally.

15. Real-World Evaluation

Measurements are kept apart by the kind of data they came from; none is combined into an overall score. Real-world audio so far measures false alarms well and recall poorly: the only real danger audio with results is 7 media clips and a partial 911 run.

Data type	Recall evidence	False-alarm evidence	Independence
A. Synthetic / generated	TTS mixes 23/30 (overlapping 1/5); generated intent test set 97.5% recall	Generated test FPR 3.6% (round 1)	Not evidence of production readiness (rule 8)
B. Prepared / recorded clips	78/84 runs (14 scenarios × 6 conditions, 5 of them simulated)	26/162 runs (UDK path only)	Regression only: tuned against
C. Real-world audio (non-Indian)	911 calls, dev partial: 194/220 calls raised any alert, inflated by dispatcher speech; full run pending	FSD50K cheering/crowd/laughter: 1–3 of ~25 each	911 sealed half untouched
D. Real Indian speech, mostly English	Real distress media 3/7	Dev YouTube (4.6 h): 3.6–4.9 speech alerts/h, scream 0–7.3/h; real media negatives 1/23; sealed YouTube (12 h): 2.2 speech/h, 2.4 scream/h, 4.6/h total	Sealed set: yes (used once)
E. Multilingual Indian speech	Not measured	Telugu recordings: 4.2/h (5.1/h in a 2026-09-27 run); FLEURS: 1–8 UDK hits per ~580 transcripts per language (ASR-level, section 6)	Dev only

The status doc's warning holds and is visible here: the scream trigger measured 0 false alarms per hour on 18 minutes of calm calibration speech, then 20–77/h (9 classes) and 2.4–18.7/h (4 classes) on real Indian YouTube audio. Small calibration sets have repeatedly overstated real-world performance.

16. Failure Taxonomy

The failures that cost recall today sit in ASR (overlap, muffling) and acoustic detection; most false alarms come from language, fuzzy/semantic matching and the scream trigger. Severity: High = a missed danger or a frequent false alarm on real audio; Medium = occasional or bounded; Low = cosmetic or theoretical. "Reproducible" = seen in more than one run or condition.

ID	Failure	Example and evidence	Component	Severity	Reproducible	Affects	Addressed?
F1	VAD / segmentation	Negations split into "I don't" + "want to go home now" (18/36 negation runs alerted); one "Please" transcribed twice	vad.py	High (was)	Yes	FP	Fixed 2026-09-25/26. Open: 8 s cap never swept; cries not flagged as speech are never analysed (not measured)
F2	ASR transcription	Two-speaker "Exact UDK_03 + UDK_09" transcribed as "You do me!"; 4/7 real distress media missed	Whisper, Parakeet	High	Yes (all 6 conditions)	Recall	No
F3	ASR hallucination	"Thanks for watching!" on traffic; "Please remember to subscribe..." on Bengali speech	Whisper (Parakeet less)	Medium	Yes	FP	Partly: confidence gate on Whisper only, n = 8 calibration

ID	Failure	Example and evidence	Component	Severity	Reproducible	Affects	Addressed?
F4	Language / code-switching	Telugu speech → "What's the name of the police?" → UDK_11; English on 95–99% of FLEURS clips	STT (English-only), no LID	High	Yes	FP; recall unknown	No (LID and Indic STT disabled)
F5	Exact-match false positive	"Third-person or quoting a story" control alerts in 6/6 conditions; one-word "back." / "go." hits on real audio	Matcher (exact)	High	Yes	FP	Partly: fragment rule; veto only when both engines agree
F6	Fuzzy false positive	"...something is going to be taken away from me" → UDK_03 (real podcast); earlier "I'm not saying..." → UDK_02 53×	Matcher (fuzzy)	Medium	Yes	FP	Partly: 75% word-coverage rule (2026-09-25)
F7	Semantic false positive	"Goodbye. Goodbye." → UDK_10; "let's go" → "Let go of me"; "It's fine, I'm safe" → "I'm not safe" (0.69)	MiniLM at 0.65	Medium	Yes	FP	Partly: veto when both engines agree; semantic ignores the fragment rule

ID	Failure	Example and evidence	Component	Severity	Reproducible	Affects	Addressed?
F8	Semantic false negative	Garbled danger ("Over please." for "Get away!") has no meaning to match	MiniLM	Medium	Yes	Recall	By design: fuzzy kept for garbled text
F9	Context interpretation	Live v1 scores "Come on now" 0.999 (cannot veto everyday fragments); v2 silenced cries after attacker speech (0.000)	mDeBERTa	High (v2), Medium (v1)	Yes	FP (v1), recall (v2)	v2 rejected; v1 limitation open
F10	Speaker attribution	No speaker information anywhere; attacker and victim words are merged	Whole pipeline	Medium	Yes	Both	Product rule: any voice saying a UDK must alert
F11	Acoustic distress	Game-audio screams 18.7/h; frightened speech scores 0.013–0.022; 28/57 real distress sounds missed	BEATs trigger	High	Yes	Both	Partly: narrowed to 4 classes; media veto and persistence rejected

ID	Failure	Example and evidence	Component	Severity	Reproducible	Affects	Addressed?
F12	Overlapping speech	UDK_03 + UDK_09 missed in every condition; TTS overlapping 1/5	STT	High	Yes	Recall	No: separation (WeSep, SepFormer) dropped for false positives
F13	Noise / muffling / clipping	Pocket condition lowest (12/14); muffled TTS 7/9	STT	Medium	Yes	Recall	Partly: Parakeet added; noise suppression rejected
F14	Threshold / calibration	Veto at 0.99 silenced "don't touch me" (p 0.24); confidence gate from n = 8; scream thresholds from 18 min of calm speech	Veto, gate, BEATs	Medium	Yes	Both	Veto moved to 0.02; others open
F15	Dataset / evaluation	Prepped tuned-on; harness skips the scream trigger; harmless-only sealed sets cannot reveal lost cries; 6 conditions of one recording counted as 6 runs	Evaluation	High	–	Both	Partly: sealed-test protocol (2026-09-28)

ID	Failure	Example and evidence	Component	Severity	Reproducible	Affects	Addressed?
F16	Integration / deployment	Retention sweep and journey eviction never called; whole-buffer VAD rescan per frame; unpinned packages and models; Denver placeholder phrase; Whisper load failure silently becomes MockSTT (no English detection); a missing frame seq stalls processing	api.py, deploy	High	Code-level	Recall and operations	No

17. Current Strengths

The system is strongest on clear English phrases under ordinary distortion and on keeping false alarms low on real English-heavy speech. Each point below is measured.

- **Distorted single-speaker English:** loud/clipped 14/14, crowd 13/14, phone 13/14 danger runs; the one miss in those conditions is the two-speaker clip.
- **Negation:** negation clips alerting fell from 18/36 to 1/36 after the VAD fix and veto; the veto flagged 0/220 held-out negations while keeping 12/12 "negation but still danger" sentences.
- **False alarms on unseen real speech:** 2.2 speech alerts per hour on 12 h of sealed YouTube audio, down from 16.0/h (conversation, dev) before the veto.
- **Complementary engines:** Parakeet recovered loud, crowd and pocket runs that broke Whisper; Whisper recovered a short quiet second-speaker phrase Parakeet dropped.

- **Conservative veto design:** it acts only where both engines heard the same words, so unclear audio can never be silenced by the text model.
- **Alert durability and security:** every frame is stored encrypted before processing; alerts go to a durable queue with deterministic IDs (no double alerts on retry), are HMAC-signed, and are retried on each heartbeat; transcripts and personal phrases are encrypted at rest; audio access needs the journey's session token and is audited.
- **Reproducible evaluation for the prepped set:** the final run logs exist and recount exactly (78/84, 26/162).

18. Current Limitations

The system is English-only, misses overlapping speakers, cannot tell who is speaking, and has operational gaps that would matter under sustained load. Each limitation is measured or read from code at 96ea038.

Area	Limitation	Evidence
Language	No Indian-language recognition; Indic speech becomes invented English	Section 6
Overlap	A UDK spoken under another voice is lost	UDK_03 + UDK_09 missed 6/6; TTS overlapping 1/5
Real distress media	3/7 caught	real_recordings
Muffled audio	Pocket is the weakest condition (12/14)	Section 14
Speaker	No speaker information; context cannot be attributed	Sections 11, 16 (F10)
Context	The live veto sees one segment (8 s or less) and cannot silence everyday fragments it scores high	Sections 4, 10
Single-engine errors	An alert heard by only one engine can never be vetoed; most real false alarms are of this kind	20/26 dev false alarms

Area	Limitation	Evidence
Scream trigger	Fires on recorded screams in media; misses frightened speech	Section 9
Custom phrase	One text-only phrase; the Denver placeholder is live when none is set; never evaluated	Section 8
Memory	Each journey's full audio stays in RAM; <code>evict_finished</code> is never called, so memory grows with every journey the process has served	<code>api.py</code> (no call site)
Storage retention	<code>retention.sweep</code> is never scheduled; encrypted audio and events on <code>/data</code> are kept indefinitely	<code>api.py</code> , <code>deploy/</code> (no call site)
CPU cost	The VAD re-scans the whole journey buffer on every frame; BEATs runs about 2 passes per second of segment audio for the scream check, plus one more	<code>vad.py</code> , <code>scream_trigger.py</code>
Silent degradation	If Whisper fails to load, English detection silently becomes MockSTT (empty transcripts), with no warning	<code>api.py</code> <code>_load_stt_backend</code>
Frame gaps	A never-arriving sequence number stalls processing for the rest of the journey	<code>api.py</code> <code>_ingest_frame</code>
Timeouts	Checked only when the journey is next accessed, not on a timer	<code>api.py</code> <code>_check_timeout</code>
Reproducibility	Python packages unpinned (except <code>onnx-asr 0.12.0</code>); Whisper, MiniLM, BEATs and Parakeet revisions unpinned	<code>deploy/requirements-deploy.txt</code> , <code>prefetch_models.py</code>
Production latency	Never measured on the c7i.2xlarge CPU	Status doc open item

19. Unresolved Technical Problems

These are open questions, stated without proposed solutions (Phase 0 rule).

1. **Overlapping speech:** how to recover a UDK spoken under another voice, when both separation models tried (WeSep, SepFormer) added false positives and wrong transcripts.
2. **Single-engine alerts:** how to reject invented English heard by one engine without also silencing a real cry heard by one engine ("You're not gonna hurt me. Get away!" was vetoed when this was tried).
3. **Real danger recall on real audio is almost unmeasured:** 7 media clips and a partial 911 run are the only real danger evidence; every larger real set is harmless-only.
4. **Recorded vs live screams:** BEATs cannot distinguish a scream in a video game or film from a live one; the one broad fix tried removed real distress media.
5. **Frightened speech without words:** it scores near zero on BEATs and on emotion2vec fear.
6. **Context without speakers:** context changes the veto's answers strongly, but without knowing who spoke it can flip a real cry to harmless (section 11).
7. **Indian-language and code-switched speech:** recall is unknown; false alarms come from invented English.
8. **Everyday fragments the veto cannot fix:** "Come on now", "Okay, let's go", "back." score high on the live model.
9. **Operations under load:** production CPU latency, memory growth per journey, and the cost of the whole-buffer VAD rescan are unmeasured.
10. **Offline vs live equivalence:** whether evaluation segmentation, engine state and the missing scream trigger change results is unverified.
11. **Custom phrases:** false-alarm and recall behaviour of personal phrases (and the placeholder) are unmeasured.

20. Reproducibility Checklist

The baseline can be rebuilt from commit 96ea038 plus one pinned private model, but not bit-for-bit: most packages and four public models are unpinned, and one audio condition uses a per-process random seed.

Item	Baseline value	Pinned?
Repository commit	github.com/sohans1092004/model @ 96ea038	Yes
Docker base	pytorch/pytorch:2.5.1-cuda12.4-cudnn9-runtime; torchaudio 2.5.1 (cu124)	Yes
Python	3.11 per the base image	NOT VERIFIED (not printed in any log)
CUDA / cuDNN	12.4 / 9 (image); production runs on CPU	Yes
Python packages	deploy/requirements-deploy.txt: only onnx-asr==0.12.0 and onnxruntime-gpu (range) pinned; faster-whisper, transformers, sentence-transformers, fastapi, rapidfuzz, webRTCvad, cryptography unpinned	No
Intent model	HF sohans1092004/udk-intent-mdeberta @ 6131acc0f5c583d3ef24b7da1f293d207f26a08b (private)	Yes
Whisper small.en	faster-whisper download by name	No
Parakeet	onnx-asr nemo-parakeet-tdt-0.6b-v2	No (library version pinned)
MiniLM	sentence-transformers paraphrase-multilingual-MiniLM-L12-v2	No
BEATs	lpepino/beats_ckpts BEATs_iter3_finetuned_on_AS2M_cpt1.pt	No
Environment variables	deploy/.env.example at 96ea038 (section 2) + UDK_DEVICE=cpu from SSM	Yes (file); SSM contents NOT VERIFIED
Input format	16 kHz mono 16-bit PCM in binary frames: >Q seq + >d client time + PCM	Yes
Output format	Event JSON (section 3 schema), webhook body identical, X-UDK-Signature HMAC-SHA256	Yes

Item	Baseline value	Pinned?
Evaluation hardware	Kaggle T4 GPU (all logged baseline runs)	–
Random seeds	Model inference is deterministic except GPU nondeterminism. <code>audio_augment.py</code> seeds the synthetic noise with <code>hash(kind)</code> , which Python randomises per process: the crowd condition uses different noise on every run unless <code>PYTHONHASHSEED</code> is fixed	No

Doc discrepancy found here: the status doc calls the crowd condition "babble noise at 10 dB SNR"; the code synthesises band-limited white noise (200–2,000 Hz) and says it is "not a claim of acoustic realism".

Baseline reproduction checklist

- ☐ Check out 96ea038 (not the local intent-context branch, whose `intent_gate.py` differs)
- ☐ Build the image with `--build-arg UDK_INTENT_MODEL_REPO=sohans1092004/udk-intent-mdeberta --build-arg UDK_INTENT_MODEL_REVISION=6131acc0...` and the hf_token secret
- ☐ Record the resolved package versions (`pip freeze`) and model revisions inside the image
- ☐ Use `deploy/.env.example`'s detection flags; set `UDK_DEVICE` explicitly
- ☐ Set `PYTHONHASHSEED` to a fixed value for any run that uses the crowd condition
- ☐ Prepped: `PREPPED_CONDITION=<normal|quiet|loud|crowd|pocket|phone>`
`PREPPED_STT_MODEL=small.en PREPPED_DEVICE=cuda UDK_ENABLE_KWS=0`
`UDK_INTENT_GATE=<model dir> UDK_INTENT_THRESHOLD=0.02 python`
`evaluate_prepped_data.py` (6 runs); expect 78/84 and 26/162
- ☐ real_recordings: `python evaluate_full_system.py` with
`UDK_EVAL_STT_MODEL=small.en, UDK_EVAL_DEVICE=cuda, UDK_ENABLE_KWS=0`
and the same intent variables; expect mixed 23/30, positives 3/7, negatives 1/23
- ☐ Real-world FA: `python evaluate_real_world.py run --root <dir> --max-min`
`<cap>` then `report`, add `--sealed` on sealed sets; the `--max-min` cap must match
(sealed run 1 used 60)
- ☐ Confirm the run log prints `intent gate ON: ... (threshold 0.02 ...)`; a missing model silently turns the gate off

21. Frozen Baseline Specification

BASELINE = GitHub main 96ea038 with deploy/.env.example's flags and the intent model at HF revision 6131acc0. Everything listed below stays fixed during Phase 1 unless a controlled experiment replaces exactly one item.

Frozen components

Component	Frozen value
Code	api.py, vad.py, stt.py, udk_engine.py, semantic.py, scream_trigger.py, beats_distress_detector.py, stt_confidence_gate.py, intent_gate.py as at 96ea038 (not the intent-context branch)
Flags	STT=1 (small.en), PARAKEET=1, KWS=0, SEMANTIC=1, STT_CONFIDENCE_GATE=1, SCREAM_DETECTION=1, SCREAM_TRIGGER=1, INTENT_THRESHOLD=0.02; LANGUAGE_ID, INDIC_STT, SEPARATION, DUAL_ASR off
Thresholds	VAD aggressiveness 1, 1.5 s pre/post-roll, 8 s cap; VERIFY 0.60, ALL 0.85; fuzzy 0.55 + 75% coverage; semantic 0.65; scream 4 classes $\geq 0.10 \times 2$ windows; corroboration 9 classes ≥ 0.30 ; veto 0.02 with agreement ≥ 80 ; repetition window 15 s
Phrase list	The 20 English UDKs in udks.py; UDK_14, 18, 20 verify-by-default
Alert protocol	TRIGGER_ALL / TRIGGER_VERIFY / NO_ACTION; event schema and deterministic event_id; HMAC-signed webhook; every alert = alarm + "Are you safe?" (website)
Evaluation sets	prepped_data (41 labelled files $\times$ 6 conditions); real_recordings (30 + 7 + 23); dev YouTube 4.6 h + Telugu 1.2 h; sealed YouTube 12 h (60 min cap per clip); 911 split by $\text{crc32}(\text{file name}) \% 2$
Harness	evaluate_prepped_data.py, evaluate_full_system.py, evaluate_real_world.py (with --sealed) at 96ea038

Baseline scorecard

Each number keeps its data type (A synthetic, B prepared clips, C real non-Indian, D real Indian English-heavy, E multilingual Indian). No overall score is given.

Dimension	Measure	Baseline	Type
ASR, English	WER	NOT MEASURED (only detection-level proxies exist)	-
ASR, Indian languages	Share of FLEURS clips returned as English text (Whisper)	95-99% across 7 languages	E
ASR, code-switching	any	NOT MEASURED	-
Recall	Prepped danger runs	78/84	B
Recall	TTS mixes	23/30	A
Recall	Real distress media	3/7	D
Recall	Real 911 calls, any alert	194/220 dev (partial, inflated by dispatcher speech)	C
False positives	Prepped control runs (UDK path only)	26/162	B
False positives	Real media negatives	1/23 files, 1/115 segments	D
False alarms/hour	Sealed YouTube 12 h: speech / scream / total	2.2 / 2.4 / 4.6	D
False alarms/hour	Dev: conversation, hard, ambient, Telugu (everything live)	5.3, 12.2, 3.9, 4.2	D, E
Robustness	Crowd (synthetic noise) / loud-clipped / pocket / phone	13/14 / 14/14 / 12/14 / 13/14	B
Robustness	Overlapping speech	1/5 TTS mixes; 0/1 recorded two-speaker scenario (all conditions)	A, B
Context	Negation clips alerting	1/36 runs	B
Context	Quote / story control	alerts in 6/6 conditions	B
Context	Harmless identical wording ("Trap", "Negative - No UDK")	alert in 5-6/6 conditions each	B

Dimension	Measure	Baseline	Type
Context	Attacker context / victim context	Not applicable to live v1 (no context); v2 probe failed on attacker context	A
Operational	GPU latency	0.57 s per prepped clip (T4, harness)	–
Operational	CPU latency (production)	NOT MEASURED; laptop CPU 1.3–1.7× audio length	–
Operational	Memory	NOT MEASURED; grows per journey (no eviction)	–
Operational	Model sizes	Parakeet about 2.4 GB, mDeBERTa about 1.1 GB; Whisper small.en, MiniLM, BEATs NOT MEASURED	–
Operational	Deployment	One Docker image, 1 uvicorn worker, EC2 c7i.2xlarge CPU, SSM secrets, private HF token at build	–

22. Recommended Experimental Protocol

Every candidate is compared against the frozen baseline on the same data, through the same harness, changing one named variable, with its pass rule written before any sealed run. This formalises the sealed-test rule adopted on 2026-09-28 and the project's recall-first ranking.

1. **One variable.** Each experiment names the single component it replaces (e.g. "the STT", "the veto model"); everything else stays as in section 21.
2. **Same harness, same data.** Run baseline and candidate through the same evaluation code. If the harness changes (e.g. adding the scream trigger to the prepped run), re-run the baseline with it first.
3. **Fixed randomness.** Fix PYTHONHASHSEED; record package versions and model revisions for both runs.

4. **Pre-registration.** Before a sealed run, commit the candidate and write its pass rule in sealed_runs.md. Share only totals from sealed sets. A set that decides a change becomes dev afterwards.
5. **Ranking.** Recall first, then false alarms, then latency. A candidate that loses any real danger detection on the regression sets fails unless the user explicitly accepts that loss.
6. **Separate by data type.** Report A-E separately (section 15); never merge datasets into one score.
7. **Harmless-only sets are not recall evidence.** Any change that can silence alerts must also be checked on danger audio (prepped, real_recordings, 911 dev, and cries after attacker speech when such data exists).
8. **New datasets.** A higher score on a new dataset is not an improvement unless the baseline is run on that same dataset and the candidate also holds on the frozen sets.
9. **Latency on production-like hardware.** Measure CPU latency, not only T4, since production runs on CPU.

Report template (one per experiment)

Measure	Baseline	Candidate	Change
Prepped danger (B)	78/84		
Prepped controls, UDK path (B)	26/162		
real_recordings: mixed / positives / negatives	23/30 / 3/7 / 1/23		
911 dev: calls alerting (C)	pending full run		
Dev false alarms/h: conversation, hard, ambient, Telugu	5.3, 12.2, 3.9, 4.2		
Sealed false alarms/h: speech / scream / total	2.2 / 2.4 / 4.6		
Latency per clip: T4 / CPU	0.57 s / not measured		
Failure cases gained (dev: clip names; sealed: count)	–		
Failure cases lost (dev: clip names; sealed: count)	–		

Measure	Baseline	Candidate	Change
Pass rule (written before the run)	-		

23. Information Missing From Current System

The biggest gaps are the live system's real behaviour (latency, memory, logs), real danger audio of any size, and anything about Indian-language speech. The current data-handling behaviour is documented first, as required, without redesigning it.

Data privacy and safety: current technical behaviour

Question	Behaviour at 96ea038
What audio is stored	Every received frame of every journey, silence included
When	On arrival, before any processing
Where	Local files under /data (S3 only if UDK_S3_* is set; not set in the deploy file)
Encryption	Fernet (MultiFernet) with a key file on the same /data volume
Raw audio retained	Yes. The retention sweep (30 days unflagged, 180 days flagged) exists but is never called, so retention is indefinite
Audio access	Signed URL valid 300 s, requires the journey's session token; every request audited
What transcript is stored	Only for alert events (the alert segment's text); other transcripts are not stored
Events	SQLite at /data (Postgres only if configured): transcript, session token and personal phrase encrypted; other fields plain
Outbound queue	Each alert, transcript included, as plaintext JSON under /data/udk_event_queue until delivered, then deleted

Question	Behaviour at 96ea038
Webhook contents	The full event JSON: journey_id, user_id, decision, UDK, layer, confidence, transcript, timestamps, model versions; HMAC-signed. TLS depends on the configured URL (NOT VERIFIED)
Speaker-identifying data	Raw voice audio is kept (as above); no speaker embeddings or voice prints are stored
Logs	Detection reasons are not logged on main (the decision-logging branch is unmerged); what uvicorn and Docker retain on the host: NOT VERIFIED

Missing information

Missing	Why it matters
Production CPU latency and memory over a real journey	Needed to judge the whole-buffer VAD rescan, BEATs cost and the missing eviction
Live logs confirming "intent gate ON" and the model versions loaded	Deployment verification (whether LaBSE also loads)
Resolved package versions and model revisions inside the live image	Reproducibility
WER of both STTs on in-domain English	ASR quality is only known through detection proxies
Real danger audio beyond 7 media clips	Recall on real audio is nearly unmeasured
Full 911 dev result and which calls raised no alert	Pending run
Any Hindi, Telugu, Hinglish or Telugu-English danger recall	Never measured
Speaker count and identity in prepped_data	Leakage assessment
Evaluation data behind the Indic STT accuracy quoted in code	Claims cannot be checked

Missing	Why it matters
Recall on cries after an attacker's words, on real audio	The case that sank v2 has no real test data
Scream-trigger false alarms on prepped controls	Harness never ran it
Custom-phrases and placeholder behaviour on real audio	Never evaluated
Website behaviour (alarm, 2-minute escalation)	Outside this repository

Inputs Required for Phase 1

Factual gaps only, no datasets named (Phase 0 rule).

- **Languages missing:** no Hindi, Telugu or other Indian-language speech with danger phrases; no measured recall in any of them.
- **Code-switching missing:** no Hinglish or Telugu-English speech, danger or harmless, with labels.
- **Real distress speech missing:** only 7 real media clips; no real (non-acted) spoken UDKs from frightened people.
- **Speaker diversity missing:** the prepped set has a handful of voices; no age, accent or gender balance is documented.
- **Overlapping speech missing:** one recorded two-speaker scenario and 5 TTS mixes.
- **Conversational context missing:** no real recordings where an attacker's words precede a victim's cry, and none of the matching harmless scenes.
- **Speaker-attributed data missing:** nothing labels who said which words.
- **Hard negatives missing:** too few real recordings of UDK words said harmlessly (quotes, stories, jokes, games, media playing nearby), especially in Indian settings.
- **Real-world Indian audio with ground truth missing:** the Indian real-world sets are harmless-only, so they measure false alarms but never recall.
- **Acoustic distress data missing:** frightened speech without screaming; live vs recorded (media) screams.
- **Device realism missing:** real phone-in-pocket, bag and phone-call audio; the current conditions are simulated (the crowd condition is filtered white noise).

- **Independent test data missing:** apart from the sealed sets, every evaluation set has been tuned against; each new batch needs a dev/sealed split before use.